

TASK 1

Network Reconnaissance Using Nmap and Wireshark

Tools Used

Tool	Purpose
Nmap	Network scanning and port discovery
Wireshark	Packet capture and traffic analysis

Install Nmap

Linux

```
sudo apt update  
sudo apt install nmap -y
```

Windows

1. Visit the official Nmap website.
 2. Download the installer.
 3. Complete installation using default settings.
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Finding Local IP Range

Linux

```
ip a
```

Windows

```
ipconfig
```

Example local network:
192.168.1.0/24

Practical Steps

Step 1: Discover Active Hosts

```
nmap -sn 192.168.1.0/24
```

Sample Output

Nmap scan report for 192.168.1.1
Host is up.

Nmap scan report for 192.168.1.5
Host is up.

Step 2: Perform TCP SYN Scan

```
sudo nmap -sS 192.168.1.0/24
```

Purpose

- Performs stealth TCP SYN scan
- Detects open TCP ports
- Faster and less detectable than full TCP connect scan

Sample Output

Nmap scan report for 192.168.1.5
Host is up.

PORT	STATE	SERVICE
22/tcp	open	ssh
80/tcp	open	http
443/tcp	open	https

Step 3: Save Scan Results

Save as Text File

```
nmap -sS 192.168.1.0/24 -oN scan_results.txt
```

Save as HTML-Compatible XML File

```
nmap -sS 192.168.1.0/24 -oX scan_results.xml
```

Convert XML to HTML:

```
xsltproc scan_results.xml -o scan_results.html
```

Wireshark Packet Analysis (Optional)

Start Wireshark

```
sudo wireshark
```

Capture Traffic

1. Select active network interface.
2. Start packet capture.
3. Run Nmap scan.
4. Observe SYN packets and responses.

Useful Wireshark Filters

Filter	Purpose
tcp	Show TCP traffic
tcp.flags.syn == 1	Show SYN packets
ip.addr == 192.168.1.5	Filter specific host
dns	Display DNS traffic

Common Ports and Services

Port	Service	Description
21	FTP	File Transfer Protocol
22	SSH	Secure remote access
23	Telnet	Remote login service
25	SMTP	Email transfer
53	DNS	Domain Name System
80	HTTP	Web traffic
110	POP3	Email retrieval
139	NetBIOS	Windows file sharing
443	HTTPS	Secure web traffic
3306	MySQL	Database service

Security Risks of Open Ports

Open Port	Potential Risk
21 (FTP)	Weak authentication, anonymous access

Open Port	Potential Risk
22 (SSH)	Brute-force login attacks
23 (Telnet)	Unencrypted credentials
80 (HTTP)	Web vulnerabilities
3306 (MySQL)	Unauthorized database access
139/445	SMB attacks and ransomware

Interview Questions and Answers

1. What is an open port?

An open port is a network port that accepts incoming connections and communicates using a specific service or protocol.

2. How does Nmap perform a TCP SYN scan?

Nmap sends a SYN packet to the target port. If the target responds with SYN-ACK, the port is considered open. Nmap then sends RST instead of completing the connection.

3. What risks are associated with open ports?

Open ports may expose vulnerable services that attackers can exploit for unauthorized access, malware delivery, or data theft.

4. Explain the difference between TCP and UDP scanning.

TCP scanning checks connection-oriented services and receives acknowledgments. UDP scanning targets connectionless services and is generally slower because many UDP services do not respond.

5. How can open ports be secured?

Unused ports should be closed, services should be updated, firewalls should restrict access, and strong authentication should be implemented.

6. What is a firewall's role regarding ports?

A firewall filters incoming and outgoing traffic and blocks unauthorized access to ports and services.

7. What is a port scan and why do attackers perform it?

A port scan identifies open ports and services on a target system. Attackers use it to discover vulnerable services and entry points.

8. How does Wireshark complement port scanning?

Wireshark captures and analyzes packets generated during scans, helping analysts understand network communication and verify scan activity.
